01 / THE OPERATING PICTURE

Data enters many specialist modules. One policy brain decides what may count.

The current system is a governed evidence network—not an authorized ranking engine. Modules emit typed evidence; the central orchestrator applies authority, applicability, evidence strength, and coverage before any output is released.

29 MODULES MAPPED
Every live manifest module appears once

14 MODULES WITH LOCKS
37 v2/v3 module-lock files

26 ACCOUNTED UNITS
2.304 GiB across v2/v3 manifests

4 UNACCOUNTED FILES
409.199 MiB raw and unactivated

HOW THE SUBMODULES INTERACT WITH THE CENTRAL BRAIN

MOLECULE 5
Identity, proposal generation, role, novelty, design gates

FORMULATION 8
Arithmetic, universe, ingredients, rules, solvency, tolerance

PERFORMANCE 4
Barrier, cure, process QC, viscosity

ASSURANCE 12
Contracts, custody, claims, replay, reporting, red-team

→

TYPED EVIDENCE BUS

PASS
FAIL
ABSTAIN
NOT_APPLICABLE

Candidate-bound provenance travels with each result.

→

CENTRAL POLICY BRAIN

optimizer-orchestrator

Reads signed module envelopes and model-card authority. Applies:

- authority ceiling
- applicability
- evidence strength
- coverage + abstention

RANK: NONE

→

AUTHORIZED OUTPUTS

Audit cohort
Reject / quarantine
Abstain
CRO packet draft

NO FINAL SCORE

MEASURED RETURN

Independent lab result → signed, candidate-bound intake → provenance ledger → eligible module evidence.

ADAPTATION RULE

Only accepted empirical returns may update an authorized method. Claim locks, held-out splits, truth labels, thresholds, and solver authority never adapt silently. Today there is no candidate-bound wet-lab feedback loop in production.

AUTHORITY LOCK INTERESTING_FOR_AUDIT is not a rank. RANK: NONE. FINAL AGGREGATE SCORE: NOT AUTHORIZED. Unknown proposals are unmade digital proposals. Dataset presence is not model validity. No safety, synthesis, procurement, qualification, performance, novelty, or execution claim is authorized.

02 / ASSURANCE + INFRASTRUCTURE · 12 MODULES

The control plane governs evidence; it does not create scientific truth.

These modules enforce identity, custody, authority, replay, claims, and packaging. Most use deterministic fixtures or ledgers—not scientific training datasets. The central brain is intentionally DISABLED as a scoring objective.

MODULE / CEILING	METHOD	DATASET / FIXTURE BINDING	WHAT IT MAY DO NOW
amphora-candidate-contract DISABLED	Typed schema and canonical identity validation	Candidate-contract parity corpus (fixture)	Deterministic control
amphora-cro-protocol-packager DISABLED	Protocol and capture-template packaging	CRO capture and falsification templates (fixture)	Packaging control
amphora-evidence-intake-validator DISABLED	Signature, hash, provenance, and candidate-binding validation	Evidence-intake rule suite	Deterministic authority gate
amphora-ip-fto-guard DISABLED	Patent and claim-category audit	patent_landscape.json	Audit only; not legal opinion
amphora-known-answer-benchmarks DISABLED	Deterministic known-answer reconciliation	BigSolDB solubility + OpenPoly counts + POINT2 run-log locks	Benchmark control
amphora-model-card-authority DISABLED	Least-privilege authority and claim-lock enforcement	Authority red-team corpus (fixture)	Policy control
amphora-optimizer-orchestrator DISABLED	Policy authorization, applicability, coverage, and abstention orchestration	Module envelopes + signed manifest; no science dataset	Central policy brain; ranking off
amphora-redteam-audit DISABLED	Deterministic adversarial claim rejection	Claim registry + negative-test corpus	Fail-closed audit
amphora-report-packager DISABLED	Claim-locked packaging and lint	Report claim-lint corpus (fixture)	Output control
amphora-run-provenance DISABLED	JCS hashing, replay, and event lineage	Fixed-seed replay corpus (fixture)	Provenance control
amphora-source-ledger DISABLED	Source, rights, and citation ledger	facts/source_records + CORR checksum lock	Custody control
amphora-sourceability-ehs DISABLED	Supplier and regulatory evidence audit	21 public supplier/regulatory documents	Evidence lookup; no safety claim

GREEN / WHAT IS STRONG

Typed contracts, content hashes, evidence custody, replay, claim locks, and fail-closed packaging.

AMBER / WHAT DATA MEANS HERE

Fixtures prove deterministic behavior. A module lock proves custody and assignment—not scientific lift.

RED / WHAT IS ABSENT

No infrastructure module may turn a weak scientific signal into rank, novelty, safety, or performance authority.

03 / FORMULATION + PERFORMANCE · 12 MODULES

Useful arithmetic and diagnostics; no validated performance objective.

Eight formulation modules and four performance modules consume a mix of public locks, local fixtures, and quarantined material. CONSTRAINT_ONLY can reject an invalid candidate; AUDIT_ONLY can draw attention; neither can rank.

MODULE / CEILING	METHOD	DATASET / FIXTURE BINDING	WHAT IT MAY DO NOW
amphora-design-rule-score AUDIT_ONLY	Directional literature rules	BigSolDB + PolyOmics + PolyVERSE + PURformance locks	Diagnostic context
amphora-envelope-gates CONSTRAINT_ONLY	Signed numeric envelope gates	Module validation fixtures	Hard constraint only
amphora-formulation-arithmetic AUDIT_ONLY	Exact mass and equivalent arithmetic	Facts + cure/corrosion verification fixtures	Arithmetic audit; no objective
amphora-formulation-universe AUDIT_ONLY	Deterministic Cartesian enumeration and accounting	Ingredient and candidate fixtures	Audit cohort construction
amphora-ingredient-library AUDIT_ONLY	Typed constant and source lookup	BigSolDB density lock + local facts	Lookup evidence
amphora-public-corrosion-benchmark AUDIT_ONLY	Grouped endpoint benchmark against baseline	CORR-DATA lock + 30 USBR public-corrosion CSVs	Diagnostic; observed no lift
amphora-solvency-score CONSTRAINT_ONLY	Carrier-solvency and phase-stability constraints	BigSolDB + OMG properties + PolyOmics + PolyVERSE locks	Hard constraint only
amphora-tolerance-robustness CONSTRAINT_ONLY	Deterministic perturbation and margin checks	BigSolDB density + RadonPy locks + local fixtures	Hard constraint only
amphora-barrier-corrosion-score AUDIT_ONLY	Measured EIS endpoint contract	CORR-DATA lock + untouched/right-quarantined ST-19168 workbook	Endpoint audit; no checked benchmark
amphora-cure-potlife-gate CONSTRAINT_ONLY	Cure, gel, film-set, and pot-life constraint checks	PolyVERSE thermoset-mechanics lock + verification fixtures	Exploratory constraint
amphora-process-qc-twin AUDIT_ONLY	Drift, missingness, and process-control diagnostics	UCI SECOM + PRONTO locks	Offline diagnostic
amphora-viscosity-score DISABLED	Rheology and sag gate	PolyVERSE melt-viscosity lock + source landscape	Model use disabled

FORMULATION

Exact arithmetic and signed envelopes can establish internal consistency. They do not establish mixability, cure quality, or coating performance.

PERFORMANCE

Public and process datasets support endpoint contracts and offline diagnostics. Public corrosion currently shows no lift versus baseline.

DECISION BOUNDARY

Independent, candidate-bound experiments remain the route to scientific authority. ST-19168 remains untouched and rights-quarantined.

04 / MOLECULE METHODS + ACCOUNTED DOWNLOADS

Molecule models are diagnostic and constraint machinery—not a ranking stack.

Five molecule modules use structural rules, exact/nearest-neighbor checks, fingerprints, ontologies, and deterministic proposal handling. No GNN, neural, or reinforcement-learning method carries authority. Potential novelty is NOT_ESTABLISHED.

MODULE / CEILING	METHOD	DATASET / FIXTURE BINDING	WHAT IT MAY DO NOW
amphora-molecule-generator AUDIT_ONLY	Deterministic proposal and quarantine pipeline	PI1M lock; OMG reaction archive is present but unactivated	Proposal audit only
amphora-molecule-property DISABLED	RDKit identity/reactive-group + exact/nearest-neighbor diagnostics	OpenPoly + OMG properties + POINT2 + PolyOmics + RadonPy + ChEBI locks; EEW/ AHEW benchmark	Model use disabled
amphora-molecule-role-ranker AUDIT_ONLY	Role ontology and role-fit benchmark	Local molecule library/role benchmark; OMG reaction archive is present but unactivated	Audit only; not ranking authority
amphora-molecule-novelty AUDIT_ONLY	Fingerprint, distance, and deduplication diagnostics	OpenPoly + PI1M + POINT2 test + PolyOmics locks	Audit only; novelty not established
amphora-molecule-design-gate CONSTRAINT_ONLY	RDKit structural applicability and hard constraints	Structural control fixtures	Hard constraint only

DOWNLOADED AND ACCOUNTED: NORMALIZED MANIFEST + NAMED MODULE LOCK

SOURCE FAMILY	NORMALIZED ACQUISITION UNITS	NAMED MODULE CONSUMERS
BigSolDB	density; solubility	design rule · ingredient library · known answer · solvency · tolerance
CORR-DATA	database; checksum	barrier · public corrosion · source ledger
OpenPoly / POINT2	counts; properties; model results; train/test; run log	known answer · molecule novelty · molecule property
PolyOmics / PolyVERSE	small molecules; χ/DMF; sorption; cohesive energy; diffusivity; melt viscosity; thermoset mechanics	design rule · molecule novelty/property · solvency · viscosity · cure
PI1M / RadonPy / OMG	structures; PI1070; physical properties	molecule generator/novelty/property · tolerance · solvency
PURformance / public process	evaluation; experiment; UCI SECOM; PRONTO	design rule · process QC
ChEBI	release 252 3-star	molecule property

26 ACQUISITION RECORDS

20 in v2 + 6 in v3; 2,473,556,779 raw bytes across manifests (2.304 GiB).

37 MODULE LOCK FILES

Locks assign accounted units to 14 modules. Several datasets intentionally serve more than one diagnostic.

IMPORTANT LIMIT

The bytes total may repeat an upstream package. “Accounted” means normalized and assigned—not trained, validated, or scoring.

05 / RAW DOWNLOADS NOT YET ACCOUNTED FOR

Four files, 409.199 MiB: present on disk, absent from every activated module binding.

“Unaccounted” is precise: raw bytes exist, but there is no normalized manifest and no module-specific lock. Catalog intent is planning metadata only; it does not mean a model currently uses the data.

RAW FILE / SHA-256	ACQUISITION UNIT	BYTES	CATALOG INTENT	WHY NOT ACCOUNTED / ACTIVATED
feedback/v1/external_data/omg_public_zips/raw/ OMG_monomers_CRU.zip `4577e389b4c33e825b2922ee714a21ee f5016ca2d8684c6355aa8d35a5da034f`	omg_structure_reaction	162,475,424	generator: TRAINING_CORPUS; role: DIAGNOSTIC; scoring false	No normalized manifest or module lock. Planned for two molecule modules; currently unused.
feedback/v1/external_data/polyuniverse_epoxy/raw/ Epoxy_1M_p.csv `9f1f5c593463b9c26a026948901e37b4 db545d17f05d5cd80baca5c3e14e0c45`	polyuniverse_pairs	265,396,937	No named consumer	No normalized manifest/module lock; upstream data license unresolved.
feedback/v1/external_data/polyuniverse_epoxy/raw/ epoxy_diE.csv `14fe4febeb0157470c113634c607bff46 f3f0b4fed6376b54e2da8315bfde448`	polyuniverse_epoxides	888,169	No named consumer	No normalized manifest/module lock; upstream data license unresolved.
feedback/v1/external_data/polyuniverse_epoxy/raw/ epoxy_diN.csv `8bdd4e9416e6805e5cbaa283acabf8a4 918f51a95d47768f0e89d05f7b960389`	polyuniverse_amines	315,528	No named consumer	No normalized manifest/module lock; upstream data license unresolved.

1 / ACTIVATE DELIBERATELY

For OMG reaction data: confirm rights + exact consumer/method, normalize, create a signed lock, then run leakage and held-out checks.

2 / HOLD POLYUNIVERSE

Keep the three PolyUniverse files quarantined until upstream licensing and derivative rights are resolved and a module owner is named.

3 / KEEP AUTHORITY CLOSED

Do not infer use from presence. Do not enable ranking until an independently authorized evidence package and movement receipt exist.

NOT COUNTED AS DOWNLOADED / UNACCOUNTED

v1 derived polymer_corpus_rows.jsonl is derived and used; PolyOmics PDF/HTML are references; the artifacts tree contains tool/model artifacts.

ORD IS MANIFEST-ONLY

feedback/v4/catalog.json describes 550 ORD objects / 1,181,061,030 bytes, but the corpus is not materialized locally. It is not counted as downloaded data.

Primary evidence: live module manifest be6d3e80…; module inventory 7bf3e74d…; v2 catalog c7ad68a0…; v3 catalog d37ce23b…; v4 catalog bea0d313…. Exact source paths and full SHA-256 values are preserved in evidence-map.json and SHA256SUMS.